

DATA SCIENCE AND MACHINE LEARNING

Day 24

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Topics

Day 22

- **Numpy
(Introduction)**



Numpy is a freely available library for performing efficient numerical computations over scalars, vectors, matrices, and more generally N-dimensional tensors

NumPy (short for Numerical Python) is a popular Python library used for scientific computing and numerical operations. It provides a powerful data structure called an array, which is faster and more memory-efficient than standard Python lists for handling large amounts of numerical data.

Key Features of NumPy:

- N-dimensional arrays (ndarray)
- Fast mathematical operations on arrays
- Linear algebra functions
- Statistical functions
- Random number generation

Some links



Official Numpy website: <https://numpy.org/>

Official documentation: <https://numpy.org/doc/stable/>

Source code: <https://github.com/numpy/numpy>

Background



Matrix Addition & Subtraction

To add two matrices, we add the corresponding elements together.
To subtract two matrices, we subtract the corresponding elements.
We can only add or subtract matrices of the same dimension.

$$\begin{bmatrix} 5 & 2 \\ 4 & 9 \\ 10 & -3 \end{bmatrix} + \begin{bmatrix} -11 & 0 \\ 7 & 1 \\ -6 & -8 \end{bmatrix} = \begin{bmatrix} 5 + (-11) & 2 + 0 \\ 4 + 7 & 9 + 1 \\ 10 + (-6) & -3 + (-8) \end{bmatrix} = \begin{bmatrix} -6 & 2 \\ 11 & 10 \\ 4 & -11 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 3 \\ 5 & 9 \\ 11 & -2 \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 8 & 1 \\ -3 & -4 \end{bmatrix} = \begin{bmatrix} 7 - (-3) & 3 - 0 \\ 5 - 8 & 9 - 1 \\ 11 - (-3) & -2 - (-4) \end{bmatrix} = \begin{bmatrix} 10 & 3 \\ -3 & 8 \\ 14 & 2 \end{bmatrix}$$

Arrays in Numpy



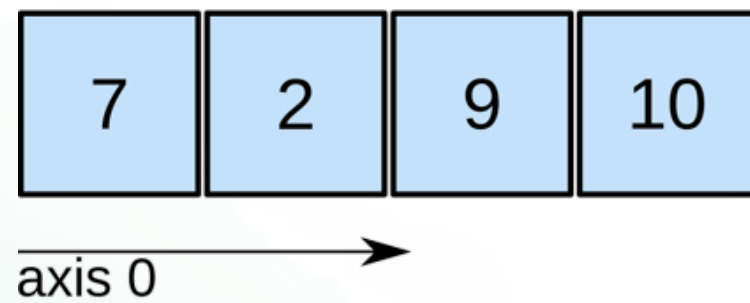
A NumPy array (ndarray) is a collection of elements of the same data type stored in a multidimensional structure. Arrays provide an efficient way to store and perform operations on numerical data.

- Number of dimensions in an array is called its rank.
- Size of the array along each dimension is called its shape.
- Elements are accessed using square brackets [] and arrays are commonly created from Python lists.

Numpy

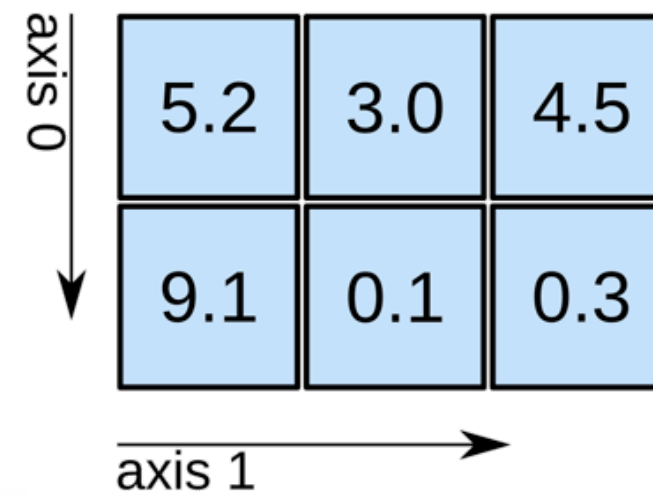


1D array



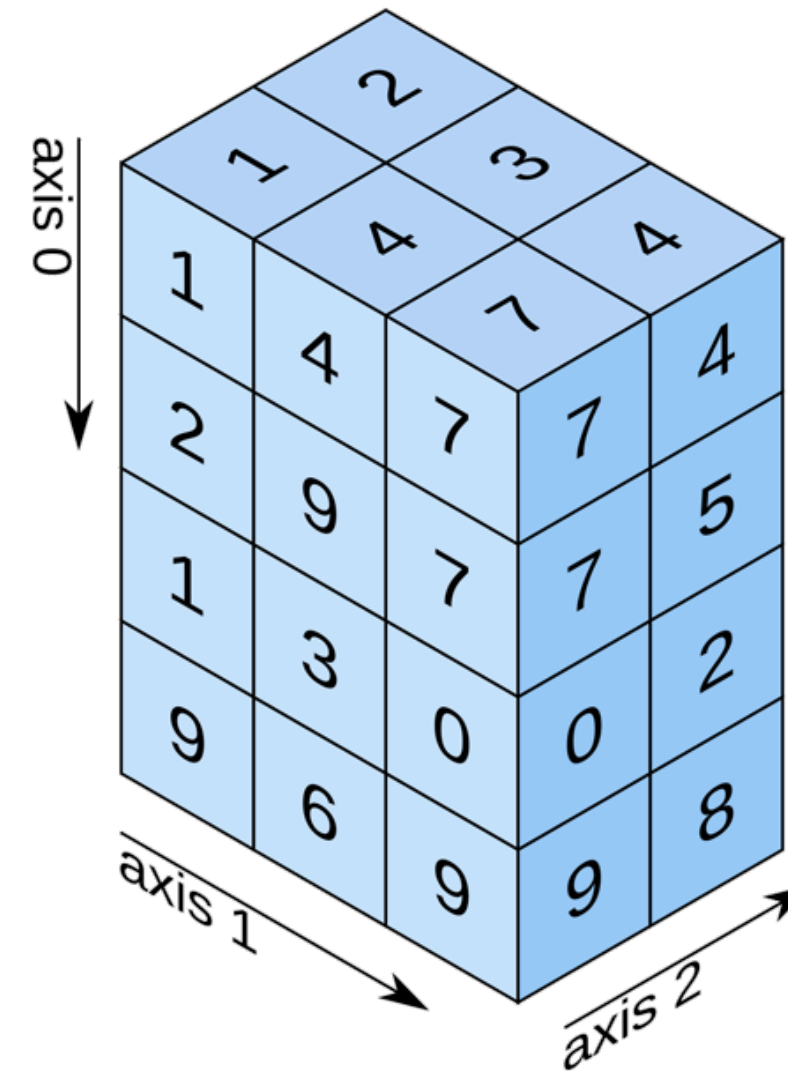
shape: (4,)

2D array



shape: (2, 3)

3D array



shape: (4, 3, 2)

Numpy



NumPy arrays can be created in several ways. Here are some of the most common methods:

- `array()` : Creates NumPy arrays from Python lists
- `zeros()` : Creates NumPy arrays with all elements initialized to 0
- `ones()` : Creates NumPy arrays with all elements initialized to 1
- `full()` : Creates an array with all elements initialized to same user-specified value
- `arange()` : Creates NumPy arrays with a sequence of values (similar to python range() function)
- `linspace()` : Creates NumPy arrays with a sequence of evenly spaced values
- `diag()` : Creates NumPy array with the provided list of numbers as the diagonal elements and zeros elsewhere
- `eye()` : Creates NumPy array with ones on the diagonal and zeros elsewhere
- `identity()` : Creates an identity matrix
- `random.rand()` : Creates NumPy array of random numbers sampled from a uniform distribution

np.zeros() and np.ones()



np.zeros() and **np.ones()** are NumPy functions used to create new arrays filled entirely with zeros (0) or ones (1).

Why Are They Needed?

1. Efficient Memory Allocation

- Quickly allocates memory for large arrays.
- Faster and more memory-efficient than Python lists.

2. Placeholders for Computations

- Used to store results that will be calculated later.
- Common in matrix operations and data processing.

3. Machine Learning Initialization

- Used to initialize weights, biases, and model parameters.
- Provides a starting point before training.

4. Boolean Masks & Matrix Operations

- Used in filtering, masking, and matrix algebra.
- Helpful in creating identity matrices and mathematical computations.

- **np.zeros()** = Buying an empty notebook with all pages blank.
- **np.ones()** = Starting with all pages marked with the value 1.



- **Add a Machine Learning & AI** Training machine learning models
 - Recommendation systems (Netflix, YouTube, Amazon)
 - Neural network calculations
- **Data Analysis**
 - Processing large datasets
 - Statistical calculations (mean, median, standard deviation)
 - Data cleaning and transformation
- **Image Processing**
 - Applying filters to images
 - Face recognition systems
 - Medical image analysis (X-rays, MRI scans)
 -
- **Financial Analysis** Stock market analysis
 - Risk assessment
 - Portfolio optimization
 - bit of body text

Class Tasks



- Create a 1D NumPy array containing the values: 10, 20, 30, 40, 50.
- Create a 2D NumPy array representing the following matrix:
 - 1 2 3
 - 4 5 6
 - 7 8 9
- Create a NumPy array of shape (3, 4) filled entirely with zeros.
- Using NumPy's `arange()` function, create an array containing odd numbers from 1 to 19.
- Using NumPy's `linspace()` function, create an array of 5 evenly spaced values between 0 and 100.

Practise Question



- Use the numpy add function to add the values 34 and 29 in the cell below.
- Declare a new array with contents [5,4,3,2,1] and slice it to select the last 3 items.
- Select all the elements in the array below excluding the last one
- Use arr as defined in 2.3. Exclude the last element from the list, but now only select every 3rd element. Remember the third index indicates stride
- You'll need to combine array comparisons and logical operators to solve this one. Find out the values in the following array that are greater than 3 AND less than 7. The output should be a boolean array.



Array Manipulation

Create and reshape arrays

`np.array()`
`reshape()`

Mathematical Functions

Perform element wise Operations

`np.add()`
`np.multiply()`

Linear Algebra

Perform matrix operations

`np.dot()`
`np.linalg.inv()`

Random Number Generation

Generate Random Data

`np.random.rand()`
`np.random.randint()`



Any Questions ?

